Control System Design Project

PID Controller Design using MATLAB

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Project Type: Control Systems / MATLAB Simulation

1. System Modeling

The system is modeled as a motor transfer function:
$G(s) = 10 / ((s+1)(s+2)(0.5s+1))$

After expansion:
$G(s) = 10 / (0.5s^3 + 2.5s^2 + 4s + 2)$

2. MATLAB Implementation

MATLAB code used to define the transfer function: num = 10; den = [0.5 2.5 4 2]; G = tf(num, den);

3. PID Controller Design

PID tuning steps:
- P controller: basic response
- PI controller: remove steady-state error
- PID controller: improve stability and reduce overshoot

MATLAB PID implementation: Kp = 1; Ki = 0.5; Kd = 0.1; C = pid(Kp,Ki,Kd); sys_cl = feedback(C*G,1); step(sys_cl);

4. Results

Results are stored in /images folder: - Step response before control - Step response after PID - System tuning comparison

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G(s)=10/(0.5s^3+2.5s^2+4s+2) ■■■■■ ■■■■■: - ■■■■■ P - ■■■■■ PI - ■■■■■ PID ■■■■■ MATLAB ■■ ■■■■ MATLAB ■■■■ ■■■■■.